GRADUATE STATISTICAL LEARNING | REVISED COMPANION

Linear Regression Explained Step by Step

Beginner intuition → advanced ESL modeling

PART I Accessible, detailed theory and formula interpretation

PART II Applied R laboratory: `Hmisc`, `rms`, `glmnet` and `pls`

Organized around *The Elements of Statistical Learning*, Chapter 3, with scaffolding from *An Introduction to Statistical Learning*, Chapter 3, and decision-theory context from ESL Chapter 2.

Design principle: Every major formula is unpacked into *what it says, why it exists, how to interpret its symbols, an illustrative example, and when it can fail.* No manual matrix construction is required in the R laboratory. The applied code fits real models and evaluates their scientific credibility.

November 2026 | Original teaching synthesis, not a reproduction of textbook pages

Reading guide and scope

This is one continuous lecture, intentionally closer in format to your supplied KNN handout than to a slide deck or a sequence of short tutoring messages. It has two parts: learn what regression is and why the methods work; then deploy them with reproducible R analyses. **The mathematical ideas and chapter ordering follow the uploaded books. The biomedical examples, explanatory bridges, and lab workflow are original teaching additions.**

Plain-language interpretation

If an equation feels unfamiliar, read it in this order: (1) identify the outcome, (2) identify what is unknown, (3) identify what the method minimizes or estimates, (4) interpret one coefficient in the units of the data, and (5) identify assumptions that permit inference or prediction. Matrix notation is optional at first reading and is translated into plain language wherever introduced.

Textbook map. ISLR Chapter 3 introduces simple/multiple regression, coefficient estimation, uncertainty, model fit, prediction, qualitative predictors, interactions, nonlinearities, diagnostics, comparisons, and a lab. ESL Chapter 3 proceeds through Sections 3.1–3.9: motivation; least squares and its statistical properties; subset selection; shrinkage (ridge, lasso, LARS); derived directions (PCR, PLS); comparison; multiple outcomes; further lasso algorithms; computation. ESL Chapter 2 underpins regression as conditional expectation under squared-error loss and introduces expected prediction error. The provided ESL selection extends through the beginning of Chapter 3 exercises; do not assume all exercises are present.

Applied example. Part II uses the 189-record historical `MASS::birthwt` dataset, predicting continuous infant birth weight (grams). It is convenient and reproducible but small, old, nonrepresentative of contemporary populations, and unsuitable for clinical deployment. The derived binary variable `low` must *never* be included as a predictor of `bwt`: that would leak the outcome into the model. The lab prespecifies a manageable model, uses bootstrap validation and nested tuning where appropriate, and distinguishes explanatory claims from predictive performance.

Package roles. `Hmisc` for descriptive information and missing-data methods; `rms` for regression strategy, nonlinear effects, contrasts and bootstrap validation; `glmnet` for authentic lasso/elastic net; `pls` for PCR/PLS. `rms::ols(penalty=)` supplies a ridge-type L_2 penalty, *not* lasso. No code builds the OLS solution via a manual design matrix. For `glmnet` and `pls`, model-interface matrix construction happens internally or through `model.matrix()` as required by those libraries; this is a software input convention, not a separate math exercise.

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PART I
Mathematical and Conceptual Lecture

1 Before any formula: what question is regression answering?

Orientation: ESL 2.4–2.9; ISLR 3.1; ESL 3.1.

1.1 An outcome, a predictor, and a scientific target

Suppose a researcher asks: *How does infant birth weight vary with maternal characteristics?* The response or outcome is Y: birth weight in grams. Predictors are quantities known or measured before the response, such as maternal age or recorded smoking status. An observation is one mother–infant record. A dataset contains many such records. To begin, we could fit a line relating birth weight to maternal age, but our goal must be stated more precisely.

A **descriptive association** summarizes how two variables move together in the observed sample. An **inferential association** estimates an unknown population relationship and acknowledges sampling uncertainty. A **prediction** uses predictors to estimate outcomes in as-yet unseen records. A **causal effect** asks what would happen to an outcome if an exposure were intervened on. These four questions are not interchangeable. Regression is a tool for the first three and may be a component of causal analysis, but the coefficient does not by itself identify a causal effect.

We use X for a predictor, Y for an outcome, x_i and y_i for measured values in individual i, and n for the number of individuals. For example, $x_7 = 25$ might mean the seventh participant is 25 years old. The *conditional mean* $E(Y \mid X = x)$ is the population average of Y among records with predictor value x. It is not the outcome for any single patient.

1.2 Why least squares? ESL's decision-theory starting point

ESL Chapter 2 asks which prediction $f(x)$ minimizes expected squared loss. Squared loss charges an error of $(y - f(x))^2$: missing by 200 grams costs four times as much as missing by 100 grams. The expected prediction error at a given x is

$$\mathbb{E}[(Y - f(x))^2 \mid X = x].$$

In words: average the squared difference between what happens and what we predict, among records with predictor value x. The prediction minimizing this conditional expectation is

$$\boxed{f^*(x) = \mathbb{E}(Y \mid X = x).}$$

Why? Write $\mu(x) = \mathbb{E}(Y \mid X = x)$ and split the prediction error into $Y - \mu(x)$ and $\mu(x) - f(x)$. When squared and averaged, the cross-term vanishes because $E(Y - \mu(x) \mid X = x) = 0$:

$$\mathbb{E}[(Y - f(x))^2 \mid X = x] = \underbrace{\mathrm{Var}(Y \mid X = x)}_{\text{outcome variability}} + \underbrace{[\mu(x) - f(x)]^2}_{\text{error in the prediction rule}}.$$

The first term cannot be reduced by choosing f; the second is smallest when $f(x) = \mu(x)$. This is the reason the *conditional mean*, not an arbitrary straight line, is the population target under squared-error loss.

Plain-language interpretation

A line is a *modeling choice* for the conditional mean. Least squares is a *fitting rule* for estimating that line. We can be excellent at minimizing squared error and still have a bad scientific model if the true conditional mean is curved, groups behave differently, the predictors are measured after the outcome, or the data are unrepresentative.

1.3 Population relationship versus fitted regression

A simple statistical regression model is

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i.$$

Read this as: an individual's outcome equals an intercept β_0, plus a slope β_1 times their predictor x_i, plus an unexplained departure ϵ_i. The model assumes $\mathbb{E}(\epsilon_i \mid X_i = x_i) = 0$ when the line represents the conditional mean. The population β_0, β_1 are unknown. We use data to estimate $\hat{\beta}_0, \hat{\beta}_1$, obtaining the fitted mean $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$. An *error* ϵ_i is defined against the unknown true model; a *residual* $e_i = y_i - \hat{y}_i$ is computed against our fitted model. They are not identical.

Worked example

Read a slope in units. Suppose a fictional fitted equation is $\widehat{\text{birthweight}} = 2400 + 18\,\text{age}$, with weight in grams and age in years. The fitted slope is *18 more grams in predicted average birth weight per additional year of age.* At age 25 the fitted value is $2400 + 18(25) = 2850$ grams. This is an association in the fitted model. It is not evidence that raising age by one year would itself cause an 18-gram increase.

1.4 Predictive versus inferential error

An error term ϵ_i reflects outcome variation not explained by included predictors. Sampling variability is different: even if the population model is fixed, another sample gives different $\hat{\beta}$. Model misspecification is different again: a line may systematically miss a curve. Finally, generalization error is the difference between the observed value and prediction on a new record. Distinguish these every time you report an error statistic.

2 ISLR foundations: how a straight line is estimated

ISLR 3.1.1–3.1.3; mathematical bridge to ESL 3.2.

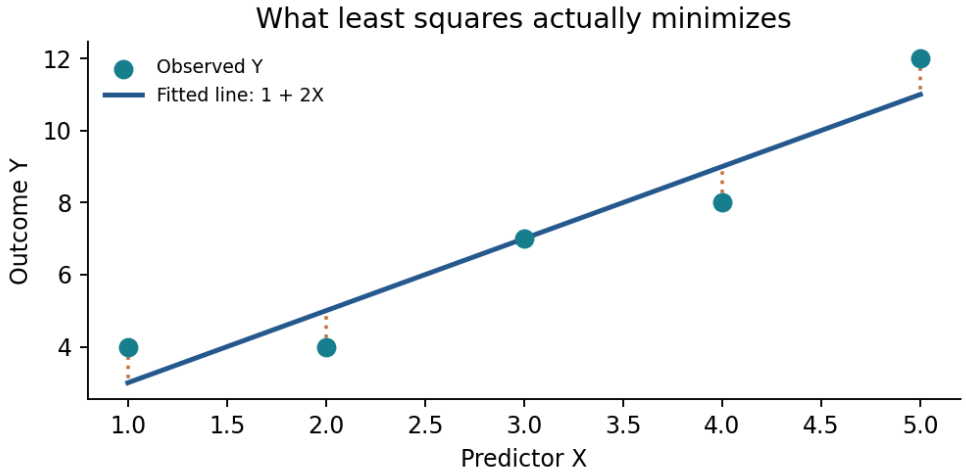
2.1 Slope, intercept, fitted value and residual

Simple linear regression has two unknown numbers. The intercept β_0 tells us the modeled mean outcome at $X = 0$. If zero falls far outside the observed range, the intercept is a mathematical anchor rather than a plausible clinical scenario. The slope β_1 represents the change in the modeled mean outcome when X rises by one unit. We choose both to bring fitted values close to the observed outcomes.

For candidate values b_0, b_1, the *residual sum of squares* is

$$\mathrm{RSS}(b_0, b_1) = \sum_{i=1}^{n} \{y_i - (b_0 + b_1 x_i)\}^2.$$

The sigma symbol $\sum$ means *add the same quantity over all individuals*. The braces contain a vertical difference: observed minus predicted. Squaring makes positive and negative differences contribute positively and magnifies large errors. OLS means choosing the numbers that produce the smallest possible RSS.



2.2 How to interpret, not memorize, the two coefficient formulas

The ordinary least-squares solution with an intercept is

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Here $\bar{x}$ is the average predictor and $\bar{y}$ is the average outcome. In the slope, the numerator asks, *when X exceeds its average, does Y also tend to exceed its average?* The denominator measures the amount of predictor variation. If all x_i are identical, the denominator is zero and the slope cannot be identified. The intercept formula ensures that the fitted line passes through the point of sample averages $(\bar{x}, \bar{y})$.

You do not need to perform derivatives every time you fit a model. The intuitive derivation is: choose an intercept and slope, sum squared residuals, and find the bottom of that bowl-shaped function. In symbols, setting each derivative of RSS to zero gives two equations whose solution is displayed above. This only requires that the predictor varies. No normality assumption is needed to compute the least-squares line.

Worked example

Five observations, every step. Let $x = (1, 2, 3, 4, 5)$ and $y = (4, 4, 7, 8, 12)$. Then $\bar{x} = 3$, $\bar{y} = 7$. The denominator is $S_{xx} = 4 + 1 + 0 + 1 + 4 = 10$. The numerator is $S_{xy} = 6 + 3 + 0 + 1 + 10 = 20$. Thus $\hat{\beta}_1 = 20/10 = 2$ and $\hat{\beta}_0 = 7 - 2(3) = 1$. Our fitted line is $\hat{y} = 1 + 2x$.
Predictions at $x = 1, 2, 3, 4, 5$ are $3, 5, 7, 9, 11$; residuals are $1, -1, 0, -1, 1$; RSS is $1+1+0+1+1 = 4$. Residuals sum to zero, as they must for OLS with an intercept. This numerical property is an algebraic result, *not* evidence that the model assumptions are true.

2.3 Why one big residual matters more than several small ones

Suppose Model A misses two observations by 2 units each and Model B misses one observation by 4 units. RSS for A is $2^2 + 2^2 = 8$, while RSS for B is $4^2 = 16$. Squared loss heavily penalizes the single large miss. This makes least squares convenient and mathematically smooth but potentially sensitive to outliers. Robust methods or careful data review may be needed if extreme values reflect miscoding or atypical data-generating processes. Do not delete observations merely because they worsen model fit.

2.4 What happens if the predictor's units change?

Suppose maternal weight is measured in pounds with coefficient 5 grams per pound. Re-express weight in kilograms: because one kilogram is about 2.205 pounds, the coefficient becomes about $5 \times 2.205 = 11.0$ grams per kilogram. Predictions remain identical if the conversion is done consistently. Coefficient magnitude therefore depends on units; a larger numerical coefficient is not necessarily a stronger scientific association. In penalized methods, standardizing predictor scales is particularly important.

3 Uncertainty: what we can and cannot claim from a fitted line

ISLR 3.1.2–3.1.3; ESL 3.2.2.

3.1 Why repeated samples give different answers

Imagine collecting many independent random samples from the same population, fitting the same prespecified model to each, and recording each slope. The slopes form a *sampling distribution*. Its mean describes bias; its spread describes uncertainty. The *standard error* (SE) is an estimate of the standard deviation of this sampling distribution. By contrast, the residual standard deviation describes how individual observed outcomes scatter around the fitted line. These measure different things.

For simple OLS define $S_{xx} = \sum_i (x_i - \bar{x})^2$. Under a correctly specified linear conditional mean with errors of zero conditional mean, common conditional variance σ^2, and errors uncorrelated across records,

$$\operatorname{Var}(\hat{\beta}_1 \mid X) = \frac{\sigma^2}{S_{xx}}, \qquad \operatorname{Var}(\hat{\beta}_0 \mid X) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right).$$

Read the first expression as: *uncertainty in a slope increases when unexplained outcome noise grows and decreases when predictors cover a wider range.* We estimate unknown σ^2 with $s^2 = \mathrm{RSS}/(n-2)$, because fitting the intercept and slope uses two degrees of freedom. Then $\widehat{SE}(\hat{\beta}_1) = s/\sqrt{S_{xx}}$. Conditional normal errors are not needed for this variance formula, but they justify exact small-sample t-inference.

Worked example

Using the five-point data, RSS=4, $n = 5$, and $S_{xx} = 10$. The residual variance estimate is $s^2 = 4/(5-2) = 4/3$. The residual SD is $s \approx 1.155$, whereas the slope's SE is $\sqrt{(4/3)/10} \approx 0.365$. Therefore, the two numerical summaries are not interchangeable. The first is about individual outcomes, the second about our estimate of the mean association.

3.2 Confidence intervals: uncertainty about a parameter

A conventional 95% confidence interval for a regression slope is

$$\boxed{\hat{\beta}_1 \ \pm \ t_{0.975,\,n-2}\,\widehat{SE}(\hat{\beta}_1).}$$

The multiplier is the 97.5th percentile of a Student t distribution with $n-2$ residual degrees of freedom, needed because the error variance is estimated. With very large samples this multiplier approaches 1.96; with very small samples it is larger. A 95% *procedure* covers the fixed true coefficient in 95% of hypothetical repeated samples if its assumptions hold. It does not mean the fixed parameter has a 95% probability of lying inside the particular interval after observing the data.

Worked example

For the five-point example, the fitted slope is 2, its SE is 0.365, and residual df=3. The 95% t-multiplier is about 3.182. Thus the interval is $2 \pm 3.182(0.365)$, or approximately $(0.838, 3.162)$. An interval based on the rough rule $\pm 2\,SE$ would be too narrow here.

3.3 Hypothesis tests: compatibility with a null claim

For a slope, the usual question is $H_0 : \beta_1 = 0$ against $H_A : \beta_1 \neq 0$. Under the classical normal-error assumptions, the test statistic is

$$t = \frac{\hat{\beta}_1 - 0}{\widehat{SE}(\hat{\beta}_1)}, \quad df = n - 2.$$

For the example, $t \approx 2/0.365 = 5.48$. A two-sided p-value is the probability, *assuming the null model*, of a test statistic at least as extreme as the one observed. It is not the probability that the null is true. A statistically significant slope may still be clinically trivial; a nonsignificant slope may still be compatible with important effects if the interval is wide. Always report estimates, intervals, units, and scientific relevance rather than just p-values.

3.4 Three numbers students often confuse: RSS, RSE, and R^2

RSS is the sum of squared residuals (squared outcome units). RSE, or the residual standard error, is $\sqrt{\mathrm{RSS}/(n-2)}$ in simple regression (outcome units). The coefficient of determination is

$$R^2 = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}, \quad \mathrm{TSS} = \sum_i (y_i - \bar{y})^2.$$

R^2 measures improvement in squared-error fit relative to predicting the sample mean, *in this sample*. It is not the probability that predictions are correct, nor does a high training R^2 guarantee external accuracy. Adding variables cannot decrease training R^2 for nested OLS models, one reason ESL insists on considering model complexity and prediction error.

3.5 Intervals for the mean versus intervals for a new patient

At a predictor value x_0, we might report a *confidence interval for the population mean* $E(Y \mid x_0)$ or a *prediction interval for a single new outcome*. The latter must account for both estimation uncertainty and the new individual's random error; it is usually wider:

$$\text{mean SE} = s\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}, \quad \text{prediction SE} = s\sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}.$$

That extra 1 is the new person's irreducible outcome variation. Both intervals become less precise as we extrapolate far from observed predictor values.

Scientific caution

A model can have a narrow confidence interval around the *average* at a specified predictor value but still make imprecise predictions for individual patients. For clinical screening decisions, this distinction is crucial. Also, nominal intervals describe the specified model under its assumptions; missing predictor information, selection effects, measurement error, and transport to another population are additional sources of uncertainty.

4 ESL 3.1–3.2: move from one predictor to several

4.1 What the multiple regression formula means

ESL Chapter 3 begins with a linear predictor in several inputs:

$$\boxed{\mathbb{E}(Y \mid X_1, \ldots, X_p) \approx \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p.}$$

Here p counts predictors, not necessarily model degrees of freedom (categorical factors and splines can use more than one coefficient). Each coefficient β_j represents the difference in the fitted conditional mean associated with a one-unit increase in X_j, *holding the other included predictors fixed*. This adjusted interpretation contrasts with an unadjusted slope from separate single-predictor models.

Worked example

Suppose the fitted equation is $\widehat{\text{birthweight}} = 2200 + 11\,\text{age} + 4\,\text{weight} - 210\,\text{smoke}$, where age is years, weight is pounds, and smoke=1 for a recorded smoking exposure. The fitted difference associated with smoke, for records with the *same included age and weight*, is −210 grams. This statement is conditional on variables in the equation. It does not mean a causal effect or that records can truly be matched on every omitted factor. If age and weight correlate, the adjusted age slope can differ sharply from a simple age-only slope.

4.2 Why correlated predictors change coefficients

Imagine that one explanatory variable predicts both another explanatory variable and the outcome. A single-predictor slope may capture both direct association and between-group differences. When the second variable is included, the first predictor's coefficient is based on its association with the outcome *after accounting for* the second. This is the heart of adjusted regression. Correlation among predictors does not automatically invalidate the model; extreme near-redundancy makes individual coefficient estimates unstable.

4.3 ESL's least-squares notation translated, one symbol at a time

ESL compresses all observed data into the matrix equation $\mathbf{y} \approx \mathbf{X}\boldsymbol{\beta}$. The boldface only denotes collections of quantities: $\mathbf{y}$ is the list of all outcomes, $\mathbf{X}$ is a table of predictor values, and $\boldsymbol{\beta}$ is the list of coefficients. When an intercept is present, the first column of $\mathbf{X}$ is a column of ones. With n records and p predictors,

$\mathbf{X}$ is $n \times (p+1)$, $\mathbf{y}$ is $n \times 1$, and $\boldsymbol{\beta}$ is $(p+1) \times 1$. Their product has the right dimension: one fitted value per record.

ESL writes

$$\text{RSS}(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

Translation: take the list of residuals and multiply each residual by itself, then add them up. The superscript T means *transpose*, or turn a column into a row; it makes the dot product legal. This is the same RSS from simple regression, not a new idea.

The least-squares answer is often written

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}.$$

Translation: a compact symbolic expression for the coefficients that minimize the total squared residuals, when the predictor columns contain enough independent information to identify a unique set of coefficients. In practice, well-designed software normally uses numerically stable decompositions rather than explicitly calculating this inverse. You do not need to implement matrix multiplication to use `rms::ols()` correctly.

Plain-language interpretation

Think of multiple regression geometrically: each predictor column points in a direction that the model can use to explain variation in the outcome. OLS chooses the nearest fitted outcome vector within the space spanned by those columns. The leftover residual vector is perpendicular to all included predictor directions. This is why OLS residuals are uncorrelated with each predictor *within the fitted sample* when the usual intercept model is used; it does not certify causal validity or independence of biological errors.

4.4 What if two predictors contain essentially the same information?

If $X_2 = 2X_1$ exactly, the model cannot uniquely separate their coefficients. More gently, if X_2 is nearly a copy of X_1, the model may estimate predictions reasonably while individual slopes vary drastically between samples. This is *collinearity*. Do not assume that dropping one variable just because it has a large p-value resolves the scientific ambiguity. Instead inspect redundancy, prespecify the model, consider effect goals, and for prediction consider shrinkage if needed.

4.5 Gauss–Markov in ordinary language

ESL 3.2.2 discusses the Gauss–Markov theorem. In a correctly specified linear conditional-mean model with errors of mean zero, equal conditional variance, and no error correlation, OLS is the *best linear unbiased estimator* (BLUE): within the class of estimators linear in the observed outcomes and unbiased under the model, it has minimal variance. It does *not* promise that OLS is always optimal under every loss function; that errors are normally distributed; that the linear model is true; or that OLS is resistant to outliers. If heteroskedasticity or dependence is present, the usual variance formula can fail even though a conditional mean association may remain meaningful.

4.6 Multiple outcomes: ESL 3.2.4

A study might record FEV1 and FVC rather than a single respiratory measurement. In multiple-response regression the outcome is a table (one column per response) and coefficients are fitted for each outcome. Ordinary separate least squares fits may give the same coefficient estimates as unrestrained multiple-output least squares, but multivariate structure can be useful when responses share information and a

scientifically meaningful joint model is imposed. This is not the same as treating repeated records from a patient as independent; within-person dependence requires an appropriate longitudinal model.

5 Model building beyond an ordinary additive line

ISLR 3.2–3.3; related ESL 3.1–3.2 modeling examples.

5.1 Are all predictors useful? The global F question

An individual t-test asks whether one adjusted coefficient equals zero in the chosen model. A *global* F-test asks whether all nonintercept coefficients equal zero together. The practical question is: does the complete set of predictors improve in-sample fit beyond predicting one constant mean? For a model with p nonintercept coefficients and n records,

$$F = \frac{(\mathrm{TSS} - \mathrm{RSS})/p}{\mathrm{RSS}/(n-p-1)}.$$

The numerator measures improvement per added coefficient. The denominator measures unexplained variation per residual degree of freedom. If the null hypothesis holds and classical assumptions apply, this ratio follows an $F_{p,n-p-1}$ distribution. If predictors are correlated, a global test can show evidence for at least one association even if many individual coefficient tests are uncertain.

More generally, a *partial* F-test compares two *nested* models, where the smaller model is contained within the larger one. Let RSS_R be RSS for the restricted model and RSS_F for the full model. If q coefficients were added,

$$F = \frac{(\mathrm{RSS}_R - \mathrm{RSS}_F)/q}{\mathrm{RSS}_F/(n-p_F-1)}.$$

The test addresses the exact set of additions, not the overall truth of either model. A data-driven model-selection process can invalidate naive post-selection p-values.

Worked example

The restricted model has RSS=600 and the full model has RSS=500. There are 100 observations and the full model has four nonintercept coefficients, so residual df=95. Two coefficients were added: $q = 2$. Then $F = [(600 - 500)/2]/(500/95) = 9.5$. This is a comparison of *two extra terms together*. Whether they are scientifically important still requires effect sizes, uncertainty and validation.

5.2 Factors and reference categories

If a predictor is categorical, such as recorded smoking (No/Yes), the model can compare the mean outcome between categories using an indicator term. For a factor with three categories, software commonly fits two coefficient indicators and treats one category as the reference. The choice of reference changes coefficient labels, not fitted values. Avoid treating category codes 1, 2, 3 as if their differences were inherently equal numerical distances. Labels and coding decisions belong in the analysis protocol.

5.3 Interactions: when a slope depends on another variable

An additive model assumes each predictor's effect on the fitted mean does not change across values of the other predictors. With an interaction,

$$\mathbb{E}(Y \mid X, Z) = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ.$$

The derivative with respect to X is $\beta_1 + \beta_3 Z$. *Translation:* the association between X and Y depends on Z. If Z is 0/1, the fitted slope of X is β_1 in group 0 and $\beta_1 + \beta_3$ in group 1. The standalone coefficient β_1 should not be described as one universal effect when an interaction is present.

Worked example

Let the fictional fitted equation be $\hat{Y} = 2000 + 20X - 100Z - 8XZ$. For $Z = 0$, the change in fitted mean per unit of X is 20. For $Z = 1$, it is $20 - 8 = 12$. The difference in slopes is -8. If one reports the main-effect coefficient 20 as the effect of X for everyone, that is wrong: it applies to the reference group $Z = 0$. We should retain both lower-order terms X and Z when fitting XZ (model hierarchy).

5.4 Why a linear regression model can contain curves

The word *linear* refers to being linear *in the unknown coefficients*, not necessarily linear in each predictor. For example,

$$\mathbb{E}(Y \mid X) = \beta_0 + \beta_1 X + \beta_2 X^2$$

is a linear regression model in its coefficients even though the mean curve bends with X. A quadratic curve imposes a particular global shape. Restricted cubic splines (also called natural cubic splines) are more flexible: they connect polynomial segments smoothly, with linear tails beyond boundary knots.



An `rms::rcs(age, 4)` term uses four knots to represent a smoothly changing association and has more than one coefficient. Do not interpret individual spline-basis coefficients as direct clinical effects. Instead inspect *predicted values and contrasts* at meaningful ages. Test nonlinearity as a *set of terms*. Specify the number of knots based on scientific expectations and sample size rather than searching until a small p-value appears.

5.5 Understanding adjusted R^2, AIC, BIC and predictive performance

Training RSS and R^2 nearly always improve as terms are added. Adjusted R^2 accounts for the number of parameters:

$$R^2_{\text{adj}} = 1 - (1 - R^2)\frac{n-1}{n-p-1}.$$

AIC and BIC trade goodness of fit against complexity under their respective likelihood-based assumptions. They answer somewhat different questions and do not automatically guarantee out-of-sample benefit. Cross-validation and bootstrap validation directly estimate predictive optimism or error when done correctly. With small samples, a flexible spline-rich model can look strong in training yet perform poorly on new records.

6 ESL 3.3: why choosing predictors is difficult

6.1 Best-subset selection: the simple idea and its combinatorial cost

ESL asks a natural question: if we start with many variables, which subset should appear in the model? Best-subset selection tries many combinations and chooses the best at each subset size according to a criterion such as RSS. With p candidate predictors there are 2^p subsets. Even with ten predictors there are 1024 possible combinations. For 30 predictors there are over a billion. More importantly, repeatedly searching for the *best* model exploits chance variation and creates selection uncertainty.

A scientific analyst should distinguish *prespecified confounders or required predictors* from *exploratory candidate predictors*. If age, site, and sex are required by the research question, forcing them into a data-driven elimination contest may undermine interpretability. For prediction, one can compare prespecified candidate strategies using resampling in which *all selection steps are repeated inside the training folds*.

6.2 Forward selection, backward elimination and forward stagewise

Forward stepwise starts with no optional predictors and adds one at a time according to a criterion. **Backward stepwise** starts with all optional predictors and removes one at a time; it needs enough observations for the initial full model. **Forward stagewise** changes coefficient values in small increments rather than adding an entire variable at once. Their computational convenience does not ensure stable selection. Correlated predictors can trade places across different samples, giving deceptively different biological stories.

Worked example

A researcher has 20 candidate biomarkers and 100 participants. One marker happens to have $p = 0.03$ in a univariable regression. Trying 20 independent null markers at the 5% level produces an expected one false positive. Selecting the smallest p-value and then treating its interval as if that model had been preplanned underestimates the selection uncertainty. This is the fundamental danger of using a stepwise search and then reporting naive intervals as confirmatory evidence.

6.3 Why the textbook compares these approaches with shrinkage

Subset selection uses a discrete decision: include a predictor or exclude it. This can be unstable when predictors contain overlapping information. Shrinkage keeps the modeling problem smoother by reducing coefficient magnitudes toward zero. Ridge usually retains all penalized coefficients, while lasso can set some exactly to zero. Neither eliminates the need to define a scientifically appropriate outcome, exclude leakage variables, and check transportability.

Scientific caution

AIC/BIC-based subset selection, forward/backward selection and lasso are useful in some settings, but they answer a *model selection or prediction* problem, not automatically a causal variable-selection problem. Do not screen covariates solely by their univariable significance when those covariates are needed for confounding control or design validity.

7 ESL 3.4: ridge regression, from intuition to formula

7.1 Why shrinkage can improve predictions

OLS tries to minimize training RSS. When predictors are highly correlated or the sample is small, OLS coefficients can fluctuate substantially. Ridge adds a penalty for large coefficient values:

$$\hat{\beta}^{\text{ridge}} = \underset{\beta_0,\boldsymbol{\beta}}{\arg\min} \left[\sum_i \left(y_i - \beta_0 - \sum_{j=1}^{p} x_{ij}\beta_j \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right].$$

Unpack the equation in plain language: choose the intercept and slopes that keep predictions close to observed outcomes, *but also* penalize very large slopes. $\arg\min$ means *the coefficient values that make the expression smallest*. The coefficient $\lambda \geq 0$ sets how expensive large slopes become. The intercept is conventionally not penalized.

What does λ do? At $\lambda = 0$, ridge is ordinary least squares (assuming a unique OLS solution). As λ grows, penalized coefficients move toward zero, trading some training fit for lower estimation variance. Ridge typically does not set coefficients exactly to zero. It can yield better future predictions even if its training RSS is slightly worse.

7.2 Why units must be addressed first

If one predictor is weight in pounds and another is age in years, a one-unit coefficient change does not mean the same thing for both. Ridge penalties on unscaled coefficients can unfairly punish variables measured on small numerical scales. Standardize predictors before comparing shrinkage paths or selecting a common penalty, or use software whose penalty matrix correctly accounts for scales. `glmnet` standardizes predictors by default, and we make its conventions explicit in the lab. A fitted coefficient on standardized inputs must be converted back to original units before comparing absolute effect sizes.

Worked example

One-coefficient ridge intuition. Suppose a centered predictor has $S_{xx} = 10$, and the unpenalized OLS slope is 2. The ridge coefficient in this special simple setup is

$$\hat{\beta}_1^{\text{ridge}} = \frac{S_{xx}}{S_{xx} + \lambda} \hat{\beta}_1^{\text{OLS}}.$$

With $\lambda = 10$, the factor is $10/(10 + 10) = 1/2$, so the ridge slope is 1. The penalty intentionally adds bias toward zero to reduce variance. This example explains the mechanism; actual software uses a precisely defined loss normalization and predictor standardization, so a numerical λ is not directly portable across packages.

7.3 What ridge buys and what it costs

Ridge is attractive when many moderately associated and correlated predictors jointly carry signal. It stabilizes coefficients but makes them biased in the ordinary unbiased-estimator sense. This is not automatically bad for prediction: expected test error includes both bias and variance, and a small amount of bias can substantially reduce variance. However, if you need interpretable etiologic coefficients, shrinkage changes their estimand-like interpretation and conventional unpenalized standard errors no longer apply unchanged.

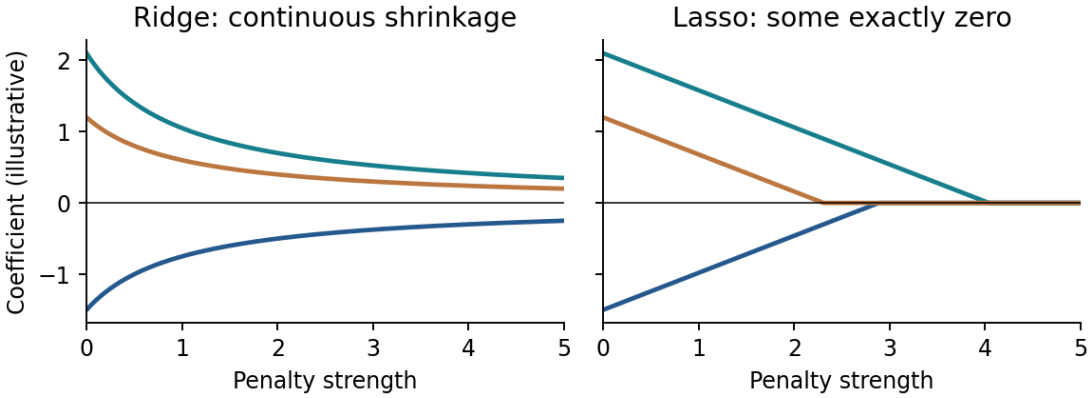
8 ESL 3.4: the lasso, sparse coefficients, and elastic net

8.1 Change the penalty and the behavior changes

The lasso solves the following optimization:

$$\hat{\beta}^{\text{lasso}} = \arg\min_{\beta_0, \boldsymbol{\beta}} \left[\sum_i \left(y_i - \beta_0 - \sum_j x_{ij} \beta_j \right)^2 + \lambda \sum_j |\beta_j| \right].$$

Compare ridge's squares β_j^2 to the lasso's absolute values $|\beta_j|$. Ridge continuously pulls coefficients toward zero. The kink in the absolute-value penalty allows lasso coefficients to become *exactly* zero. Lasso therefore performs both shrinkage and variable selection, especially when the tuning penalty is appreciable.



8.2 Why correlated predictors create a problem for interpretation

When two biomarkers are almost duplicates, a lasso fit may select one and discard the other, while a resampled dataset selects the opposite. The single selected marker should not be presented as uniquely causal, mechanistically privileged, or even stably predictive without additional evidence. Ridge tends to share signal across correlated predictors. **Elastic net** combines the two penalties:

$$\text{loss} + \lambda \left[\alpha \sum_j |\beta_j| + \frac{1-\alpha}{2} \sum_j \beta_j^2 \right],$$

where $\alpha = 1$ is lasso and $\alpha = 0$ is ridge in `glmnet`'s convention. Intermediate α values often give more stable behavior with strongly correlated predictors. *Note:* this objective is written schematically; `glmnet` scales the squared-error loss by $1/(2n)$, so numerical values of λ depend on software convention.

8.3 Choosing the tuning penalty scientifically

The penalty is not chosen because it makes a particular variable significant or produces a preferred story. Instead, train candidate models across a range of penalties and estimate prediction error by cross-validation. The frequently reported λ_{min} minimizes cross-validated error. λ_{1se} is the largest penalty whose error is within approximately one estimated standard error of the minimum; it generally favors simpler models, but the error estimate itself is noisy. Choose the penalty *inside* every external resampling split used to evaluate predictive performance.

Worked example

Suppose fivefold CV gives estimated RMSE of 515 grams at λ_{min} and 523 grams at λ_{1se}. It may be reasonable to prefer the latter if the science favors simplicity and the difference is unimportant for the use case. Do not interpret these as proven differences without addressing sampling variability, particularly in a small study.

8.4 Least angle regression: the ESL algorithmic bridge

ESL introduces least angle regression (LAR) and least angle regression with lasso modifications (LARS). Think of LARS as an efficient way to trace how coefficients enter and move as model complexity increases; it is related to forward stagewise fitting and the piecewise-linear coefficient paths of lasso. For applied work, `glmnet` uses coordinate-descent algorithms and is usually the simplest reproducible deployment choice. Understanding the reason for a *path* matters more here than deriving the update rule.

9 ESL 3.5: principal components regression and PLS

9.1 The problem: many predictors with overlapping information

Suppose 20 laboratory measures are highly correlated because they reflect a shared physiological process. Fitting all 20 separately can make OLS coefficients unstable. One alternative is to create *new combined directions* from the predictor information and regress the outcome on a smaller set of those directions. ESL discusses two major approaches.

9.2 Principal components regression (PCR)

Principal components analysis (PCA) finds mutually orthogonal directions capturing decreasing amounts of variability in the predictors X *without looking at the outcome* Y. PCR then fits regression to the first M selected components. More components generally improve training fit, but not always test prediction. The critical caveat is that the direction with the greatest variance in X need not be the direction most related to Y. Therefore, PCR can discard predictive information while retaining high-variance nuisance variation.

In one sentence: **PCR keeps directions that explain the predictors well; it does not directly optimize predictive relevance to the response.**

9.3 Partial least squares (PLS)

PLS constructs directions using both predictor variation and association with the outcome. Its directions are chosen to capture combinations of X that are useful for explaining Y, in a sequential supervised construction. PLS may outperform PCR when low-variance predictor directions are strongly related to outcome, but it can still overfit and requires component selection by resampling. Do not assume that PLS is automatically better than PCR or regularized regression.

Worked example

Imagine two standardized biomarkers, X_1 and X_2. Their common average varies greatly but is unrelated to the outcome; their small difference predicts outcome well. PCA may prioritize the average because it varies most. PCR keeping only one component would miss the useful difference. PLS, because it considers the outcome, has a route to prioritize the useful difference. Whether that advantage survives in new patients must be validated.

9.4 PCR, PLS, ridge and lasso compared

Method	What it changes	Main caution
OLS	Fits each coefficient without shrinkage	Unstable with strong collinearity or high complexity
Ridge	Shrinks all included coefficients	Doesn't usually select a sparse subset
Lasso	Shrinks and may zero some coefficients	Variable selection can be unstable
PCR	Regresses on high-variance directions in predictors	High variance is not the same as prediction signal
PLS	Regresses on directions linked to predictors and outcome	Supervision can overfit; tune components

10 ESL 3.6–3.9: extensions without losing the central idea

10.1 How ESL compares shrinkage and selection strategies (3.6)

ESL contrasts their behavior under different predictor correlations and signal patterns. There is no universal winner. A few strong signals may favor sparsity; many distributed weak signals may favor ridge; group-like correlation can favor elastic net. What wins on one sample may lose on another. All should be compared using a consistent prediction target and a fair resampling plan rather than in-sample R^2.

10.2 Multiple-outcome selection and shrinkage (3.7)

When outcomes are a related group (e.g., several correlated pulmonary function measurements), penalties can be imposed jointly rather than fitting entirely unrelated sets of coefficients. The motivation is to borrow information or select predictor patterns shared across outcomes. This goes beyond ordinary univariate OLS and requires a clearly defined response structure and appropriate software. It is not an excuse to ignore differences in measurement scale or clinical meaning between outcomes.

10.3 Lasso-related path methods (3.8)

ESL discusses incremental forward stagewise updates, piecewise-linear regularization paths, the Dantzig selector, the grouped lasso, and pathwise coordinate optimization. They share a theme: efficient constrained fitting of many related candidate solutions. Group lasso can penalize entire groups of coefficients together, useful when a categorical variable or spline produces multiple columns that should be treated as one scientific term. Standard lasso instead may choose isolated columns from a group, complicating interpretation. The Dantzig selector uses a different constraint based on correlations between predictors and residuals; it should not be presented as identical to the lasso even when some results coincide.

10.4 Computational considerations (3.9)

Ordinary least squares is fast for moderate-size models. Exhaustive subset selection becomes expensive as the number of predictors grows. Regularization paths are often efficiently computed through specialized algorithms. Practical choices also depend on missingness, sparse inputs, feature scaling, and cross-validation cost. Scientifically defensible modeling is a combination of the statistical objective, valid input data, appropriate fitting algorithms, and honest validation.

11 Model diagnostics: when the attractive regression result is unreliable

ISLR 3.3–3.4 and a supplementary advanced regression strategy lens.

11.1 Residual plots are questions, not certificates

Plot residuals against fitted values and against key continuous predictors. Random scatter around zero is consistent with—but does not prove—a reasonable mean model. Curvature suggests unmodeled nonlinear structure. A funnel shape suggests heteroskedasticity. Distinct groups may signal interactions, omitted categorical variables, or site effects. A temporal pattern may signal correlated errors. Each pattern suggests a different response; automatically throwing all data into a flexible model is not a principled fix.

11.2 Leverage, outliers and influential observations

A large *residual* means the observed response is far from its fitted value. High *leverage* means a predictor pattern is unusual in the sample. An *influential* observation materially changes the fitted result when removed or perturbed. These three ideas overlap but are not identical. In a biomedical dataset, review values for transcription errors, genuinely rare patient subgroups and model misspecification before deciding how to handle them. Sensitivity analyses should be planned and reported, not used to select a preferred conclusion.

11.3 Heteroskedasticity, dependence and normality

Under unequal error variance, OLS can still estimate an average linear relationship, but ordinary homoskedastic standard errors may be wrong. Heteroskedasticity-consistent covariance estimates are one possible remedy for inference. With repeated measurements within patient or clustering within sites, use a dependence-aware method (e.g., cluster-robust methods, mixed models, or GLS) selected for the scientific design. Normality of residuals is most relevant for exact small-sample inferential formulas; it is not a universal requirement for fitting a linear mean with least squares.

11.4 Missing data is not a minor cleaning step

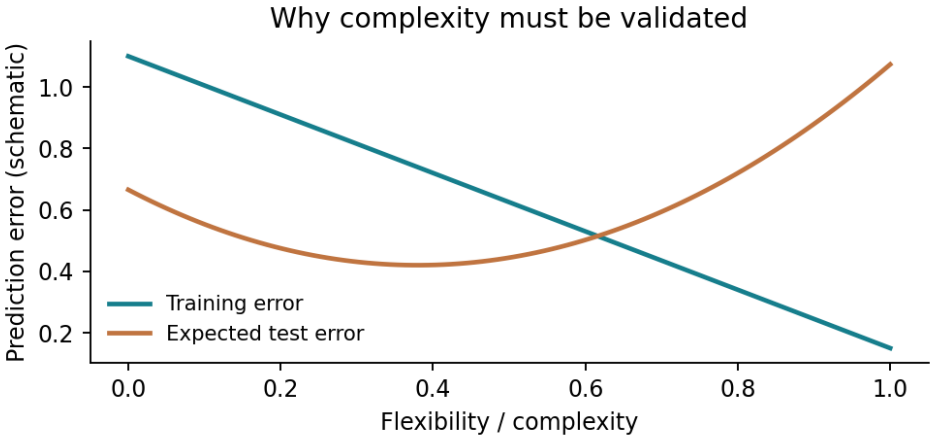
Complete-case analysis changes the analysis sample. If missingness relates to the outcome or covariates, it may create bias or reduce generalizability. Multiple imputation attempts to fill in missing predictors with uncertainty while respecting relationships among variables. It requires assumptions such as missing-at-random given the imputation predictors and a well-specified model. Imputation must be integrated with the validation process; performing it on the full dataset *before splitting for external evaluation* can leak information. The lab includes an *explicitly artificial missingness demonstration* rather than claiming that an inherently complete teaching dataset has real missing values.

12 A unifying model-selection and validation strategy

ESL 2.9; ESL 3.3–3.6; applied regression strategy supplementary.

12.1 Bias–variance tradeoff is about expected future error

Prediction error at a new point can be decomposed schematically into irreducible noise, squared bias of the fitted prediction, and variance across training samples. Very rigid models can miss curvature (high bias). Extremely flexible models can follow noise (high variance). Training error alone generally falls as complexity increases; expected test error can be U-shaped. This is the rationale for regularization and for choosing model complexity using properly separated data or resampling.



12.2 Three separate forms of validation

Internal validation estimates optimism under the same data-generating setting using bootstrap or cross-validation. **Temporal/external validation** asks how the model works in data from another time, site or system. **Clinical utility evaluation** asks whether predictions support better decisions than alternatives at realistic thresholds and costs. A model can be well calibrated internally and still fail when coding definitions, disease prevalence or practice patterns change.

12.3 What must stay inside resampling?

Anything learned from data—including missing-data imputation rules, predictor scaling, feature screening, variable selection, spline flexibility if tuned, and the lasso penalty—belongs inside each training fold or bootstrap refitting step when estimating out-of-sample performance. If you use all records to choose a variable set and only then cross-validate the coefficients, the resulting error estimate is optimistic. A nested CV scheme has an *inner loop* for tuning and an *outer loop* for evaluating the complete modeling strategy.

12.4 End-of-lecture integrative checklist

Before opening R, identify the target Y, candidate predictors and their units, sample size, missingness, leakage risks, whether the scientific goal is association or prediction, plausible nonlinearities/interactions, an inferential assumption audit, and a validation plan. Then decide whether the simplest defensible OLS model is enough or whether the evidence justifies shrinkage, variable selection or derived directions. ESL's central lesson is not that modern algorithms always beat OLS: it is that assumptions, complexity and out-of-sample evaluation determine when each method is appropriate.

13 ESL formula translator: read the notation without getting lost

A guided glossary of central ideas from ESL 3.2–3.8; not a substitute for the chapter's proofs.

The goal of this section is to remove the common jump from a simple clinical question to an abstract equation. For each expression, ask *what is observed, what is unknown, what is optimized, and what changes when I fit or tune the model?*

13.1 RSS: the score that OLS minimizes

$$\text{RSS}(\beta) = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \qquad \text{with} \qquad \hat{y}_i = \beta_0 + \sum_{j=1}^{p} x_{ij}\beta_j.$$

Observed: the outcome y_i and predictor values x_{ij}. **Unknown:** coefficients β_j. **Optimized:** the total squared difference between observed and modeled outcomes. **Interpretation:** less RSS is better *on the fitting sample*, but may not be better in a new sample. When adding a predictor, RSS cannot increase in ordinary nested OLS because the old model remains a permissible special case of the new model.

13.2 The normal equations: what software solves internally

$$\mathbf{X}^T(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}.$$

The left-hand side is the collection of correlations in a dot-product sense between each predictor column and the fitted residual vector. At the least-squares solution, none of the included directions can further reduce RSS by a small movement in that coefficient. *This explains why residuals are perpendicular to fitted predictor directions in sample.* It does not show that the true biological errors are independent of predictors.

13.3 The projection: how a fitted model finds a nearest point

$$\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}, \quad \mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T.$$

H is called the hat or projection matrix because it converts **y** into $\hat{\mathbf{y}}$. The fitted outcome vector is the nearest point to the observed outcome vector in the model's predictor space under Euclidean squared distance. The diagonal values h_{ii} describe leverage. A record with an unusual predictor combination can have high leverage even when its residual is small. There is no need to calculate **H** manually in the applied lab; use `hatvalues()` for diagnosis.

13.4 Conditional coefficient variance: how information affects precision

$$\text{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}.$$

A variance-covariance matrix is a table: the diagonal entries estimate uncertainties of individual coefficients; off-diagonal entries describe how estimates fluctuate together. More information in the predictor patterns generally improves precision, but high collinearity makes the inverse unstable. This formula assumes zero-mean, homoskedastic, uncorrelated model errors given the predictors. Real EHR data may violate those assumptions through clustering, varying precision or model misspecification.

13.5 Gauss–Markov: the limited meaning of *best*

$$\text{Var}(\tilde{\boldsymbol{\beta}} \mid \mathbf{X}) - \text{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) \succeq 0$$

for a competing linear unbiased estimator $\tilde{\boldsymbol{\beta}}$ under the theorem's assumptions. The symbol $\succeq 0$ means the variance difference is positive semidefinite, i.e., no linear contrast is more precisely estimated by the competitor in that class. *Best* does not mean best predictor under all distributions and tuning strategies, nor does it say OLS solves confounding.

13.6 Partial regression: what *adjusted* really removes

ESL's multiple-regression discussion can be understood by temporarily removing from X_1 what is explained by all the other predictors, then removing from Y what is explained by those same predictors. Regressing the leftover part of Y on the leftover part of X_1 yields the same adjusted slope as a full OLS model. This is the Frisch–Waugh–Lovell idea. It explains why a large unadjusted association can shrink after adjustment and why highly correlated variables leave little independent variation for separate coefficient estimation. It is a computational identity, not an unconfoundedness guarantee.

13.7 Ridge shrinkage factor: why strong directions shrink less

For a simplified centered design expressed in orthogonal directions, a ridge coefficient in each direction is multiplied by a factor resembling

$$\frac{d_j^2}{d_j^2 + \lambda},$$

where d_j measures strength of information in predictor direction j. If d_j^2 is large compared with λ, the factor is near one and the direction is scarcely shrunk. If d_j^2 is small, shrinkage is strong. The formula explains why ridge stabilizes poorly informed directions; it is not an instruction to calculate a singular value decomposition before fitting your applied model.

13.8 Effective degrees of freedom: how complex is penalized regression?

$$\mathrm{df}(\lambda) = \mathrm{tr}(\mathbf{H}_\lambda).$$

tr means add the diagonal entries of a matrix. The resulting quantity measures the effective flexibility of a fitted smoother or penalized linear estimator. For a ridge fit, effective df generally decreases as the penalty increases, even though many coefficients remain nonzero. The intercept is conventionally handled separately. This is why counting nonzero coefficients is not a valid complexity measure for ridge.

13.9 Lasso's constraint: what a sparse result actually means

$$\sum_{j=1}^{p} |\beta_j| \le t.$$

The lasso optimization can be viewed as minimizing squared error subject to a limited total absolute coefficient budget t. A small budget forces competition among predictors. Under certain correlation patterns, some coefficients become exactly zero. This is an *algorithmic variable-selection mechanism*, not evidence that discarded variables have no association or that selected variables are true causes.

13.10 Cross-validation error: estimating loss on records not used to fit

$$\widehat{\mathrm{CV}}(\lambda) = \frac{1}{K} \sum_{k=1}^{K} \frac{1}{|V_k|} \sum_{i \in V_k} \{y_i - \hat{f}_{-k,\lambda}(x_i)\}^2.$$

V_k is validation fold k. $\hat{f}_{-k,\lambda}$ means the model is fitted *without* that fold, using penalty λ. The inner sum scores errors on the held-out fold; the outer average combines folds. If folds are unequal in size, compute a weighted average of individual losses rather than naively averaging fold-specific means. The important scientific requirement is that all preprocessing and tuning learn only from the training portion relevant to each prediction. When CV chooses λ, an additional outer evaluation is desirable for an unbiased estimate of the selected workflow's performance.

13.11 Expected test error: the bias–variance equation

For a new response $Y_0 = f(x_0) + \epsilon_0$, the expected squared prediction error can be decomposed as

$$\mathbb{E}[(Y_0 - \hat{f}(x_0))^2] = \sigma^2 + \text{Bias}\{\hat{f}(x_0)\}^2 + \text{Var}\{\hat{f}(x_0)\},$$

under the usual independent, mean-zero test-noise setup. The σ^2 term is irreducible noise. The squared-bias term captures systematic model error, and the variance term captures sample-to-sample fluctuations of fitted predictions. Ridge and lasso intentionally accept bias if the reduction in variance can improve overall test error. This formula describes a prediction target, not uncertainty in a treatment effect.

13.12 PCR and PLS: what exactly is a component?

A component is a weighted combination $z = Xw$ of measured predictors, where w describes how strongly each predictor contributes to the new direction. PCR chooses w to capture variance of the predictors; PLS uses outcome information to help choose relevant directions. Both then regress Y on a smaller number of components. The number retained is a model-complexity decision requiring validation. Component weights are not automatically interpretable biomedical entities.

14 Guided case study I: from observations to adjusted associations

A longer explanatory bridge for the gap between ISLR 3.1 and ESL 3.2.

14.1 Start with the scientific unit, not the regression command

Suppose a health system has repeated clinical encounters for some patients. If the outcome is a measurement from each encounter, the unit of analysis could be an encounter, whereas the independent sampling unit may still be the patient. That matters: conventional OLS treats errors as uncorrelated across observations unless a dependence-aware procedure is specified. If a patient contributes ten measurements and another contributes one, a naive encounter-level model effectively gives the first patient more weight. A carefully defined analysis might instead summarize one measurement per patient, model within-person change, or use a repeated-measures framework. The regression equation alone does not decide the correct unit.

In our `birthwt` teaching dataset, the unit is one pregnancy record. Our outcome is recorded birth weight in grams. Suppose the fitted model includes maternal age, weight, smoking and a categorical race variable. Before interpreting the result, ask whether each predictor was recorded before the birth-weight outcome, whether it has a clear operational definition, and whether the covariate could itself be changed by or derived from the outcome. The binary `low` indicator fails the last test because it is computed from birth weight.

14.2 Why the same association can change after adjustment

Think of two groups of patients, A and B. Within each group, older individuals could have a weak positive relationship between X and Y, but group B could be older on average and have a much lower outcome for an unrelated reason. When all records are pooled, the unadjusted slope may even be negative. Adding a group indicator separates within-group differences in X from differences in group baseline outcome. This is a version of Simpson's paradox: pooling can hide or reverse group-specific associations.

Worked example

Consider four illustrative pairs. In group A, $(X, Y) = (1, 5), (2, 6)$; in group B, $(X, Y) = (4, 1), (5, 2)$. Within each group, a one-unit rise in X goes with a one-unit rise in Y. But group B has much larger X and much smaller outcomes. A single pooled line fitted without group adjustment has a negative slope. A regression with a group intercept can recover the positive within-group association. The lesson is *not* that adjustment always moves a coefficient toward the truth: the set of adjustment variables must be justified by the scientific question and causal/design structure.

14.3 Why a factor does not get one coefficient per category in ordinary coding

Suppose `race` has three recorded categories. A regression containing an intercept cannot also have three fully independent dummy columns plus an intercept, because knowing two dummies determines the third: their sum equals one. Ordinary reference coding uses two indicators, with one reference group absorbed into the intercept. The coefficient for `raceBlack` describes a difference from the reference category, at fixed values of other included variables. If you change the reference, the two coefficients change, but fitted outcomes do not. This is an identifiability and interpretation issue, not a flaw in the software.

14.4 What the word adjusted does not guarantee

An adjusted coefficient holds *modeled* covariates fixed, not all biological, social or clinical determinants. If important shared causes of exposure and outcome are omitted, an adjusted estimate can still be confounded. If adjustment includes a consequence of the exposure, mediator, collider, or variable measured after the outcome, it can produce a misleading quantity. A regression model is therefore only one component of research design. The broad goal of ESL Chapter 3 is to estimate or predict relationships in input space, not to supply an automatic causal identification strategy.

15 Guided case study II: three types of uncertainty in one example

Scaffolding for ISLR 3.1.2–3.2 and ESL 2.9.

15.1 Parameter uncertainty

Suppose one study estimates a maternal-age coefficient of 10 grams/year with standard error 2. Another study estimates the same slope but with standard error 15. Although point estimates are identical, the second study is far less precise. A confidence interval summarizes that precision under its inferential assumptions. Larger n, less unexplained variation, and more informative predictor ranges often reduce standard errors; dependence, collinearity and sparse subgroups can have the opposite effect.

15.2 Individual prediction uncertainty

Even if a huge sample estimates the population mean very accurately, birth weight at age 30 is not one number shared by every person. Many unmeasured determinants remain. The uncertainty in predicting a new individual includes both the population-mean estimation uncertainty and residual outcome variation. This is why prediction intervals are wider than confidence intervals for a mean, sometimes dramatically so. A statistically strong maternal-age coefficient does not imply a high-quality individual-risk prediction

tool.

15.3 Uncertainty about the model itself

Both types above usually assume the chosen model form is the one being evaluated. But perhaps the relationship is curved, perhaps age interacts with another predictor, or perhaps data collection has changed across sites. This is *model uncertainty*. Standard textbook OLS intervals for a fixed prespecified model do not automatically include extra uncertainty from searching dozens of possible specifications. Internal validation, sensitivity analyses, external validation and careful theory are needed to assess this uncertainty.

Worked example

Suppose Model A uses age alone and achieves training RMSE 520 grams. Model B uses age, weight, smoking, a six-knot age spline and several interactions; it achieves training RMSE 460 grams. One may be tempted to call Model B superior. But B has much more flexibility. If B's internal validation RMSE is 610 grams whereas A's is 535 grams, A may be the better predictive candidate despite its worse training fit. This illustrates why *fit quality* and *future prediction quality* are distinct quantities.

16 Guided case study III: comparing two regression models correctly

16.1 Three kinds of model comparisons

First, compare *nested inferential models* with a partial F-test if classical assumptions are reasonable, answering whether a defined set of extra predictors improves the fitted mean. Second, compare *non-nested descriptive models* using the same observations and a suitable fit/complexity criterion when scientifically appropriate. Third, compare *predictive strategies* using outer-fold, bootstrap or external performance. These comparisons answer different questions. Do not use a training-set p-value to proclaim an algorithm's future predictive superiority.

16.2 A fully worked nested-model test

Assume a restricted model has two nonintercept coefficients and RSS=1000, while an expanded model adds two more coefficients and reduces RSS to 940. There are 80 records. The expanded model has residual degrees of freedom $80-4-1=75$. The improvement per added coefficient is $(1000-940)/2=30$. The remaining error per residual degree of freedom is $940/75 \approx 12.533$. The partial F statistic is $30/12.533 \approx 2.394$ on $(2, 75)$ degrees of freedom. This tests whether the added two coefficients are jointly zero under the model; it does not mean the expanded model predicts better on a new site.

16.3 Why an interaction changes how you report a main coefficient

Suppose the fitted model is

$$\hat{Y} = 2500 + 15A - 100S - 5AS,$$

where A is age centered at 25 and $S=1$ for recorded smoking. The fitted difference between smokers and nonsmokers is $-100-5A$. At age 25, $A=0$, and the adjusted difference is -100 grams; at age 35, $A=10$, it is -150 grams. Centering age did not change model predictions; it changed the meaning of the

main effect of smoking to a meaningful reference age. This is a useful implementation strategy whenever interactions are scientifically justified.

Plain-language interpretation

A small p-value for a model term is not a complete scientific interpretation. For a spline or interaction, show predicted contrasts over an explicit range, state the covariate values at which those contrasts apply, and report uncertainty. A curve can change direction, and an interaction can reverse an association across groups; one coefficient cannot summarize them honestly.

17 Guided case study IV: why ridge and lasso can disagree

17.1 A loss-and-penalty comparison with actual numbers

Consider two hypothetical fitted candidates. Model A has RSS=100 and coefficient values $(3, 0)$. Model B has RSS=104 and coefficient values $(1.5, 1.5)$. Without a penalty, A wins because 100 is less than 104. With a ridge penalty $\lambda = 2$, the scores are:

$$\begin{aligned} \text{Ridge A:} \quad & 100 + 2(3^2 + 0^2) = 118, \\ \text{Ridge B:} \quad & 104 + 2(1.5^2 + 1.5^2) = 113. \end{aligned}$$

Under this illustrative ridge objective, B is preferred despite a slightly worse RSS, because the squared-coefficient penalty rewards spreading magnitude across the two slopes. With an L_1 lasso penalty of $\lambda = 2$, the scores are:

$$\begin{aligned} \text{Lasso A:} \quad & 100 + 2(|3| + |0|) = 106, \\ \text{Lasso B:} \quad & 104 + 2(|1.5| + |1.5|) = 110. \end{aligned}$$

Lasso prefers A among these two possibilities. The example does not prove global optimization for either problem; it makes clear that *changing the penalty changes the preference among candidate models.* Numerical penalties have different normalizations in different software packages.

17.2 Why coefficient selection is not a discovery claim

If X_1 and X_2 are two nearly duplicate biomarkers, selecting one in lasso merely says the chosen predictive optimization could obtain adequate performance without the other at the selected penalty. Different folds can swap them. It does not show that the selected biomarker is causally upstream or that the unselected one has no biological association. In a GWAS setting with LD-correlated variants, this issue becomes especially consequential: regularization selection is not synonymous with fine-mapping or causal-variant identification.

17.3 Why ridge is not automatically the better scientific model

Ridge helps reduce unstable coefficients when many correlated predictors carry shared signal. But if the scientific question asks for a specific covariate's adjusted mean association, ridge changes the quantity estimated by pulling all penalized coefficients toward zero. Likewise, if a strong nonlinear effect is omitted from the model, ridge cannot reconstruct it merely by shrinking coefficients. The best procedure depends on the estimand, the role of the variables, the form of the relationship and the amount of available information.

18 Guided case study V: PCR, PLS and the difference between variance and signal

18.1 A thought experiment with two biomarkers

Let X_1 and X_2 be standardized biomarkers that usually rise and fall together. Their *average* $(X_1 + X_2)/2$ varies a great deal between patients. Their *difference* $X_1 - X_2$ varies only a little. Suppose, however, that disease biology connects the outcome to the difference and not to the average. PCA prioritizes the high-variance average direction. If PCR retains only one component, it misses the clinically relevant contrast. PLS may identify the contrast because it uses the association with the response. In a tiny sample, though, this may reflect accidental correlations and will not necessarily generalize.

18.2 What M, the number of components, means

When PCR uses $M = 1$, it fits one derived predictor; at $M = 2$, two; and so on. This provides a continuous sequence of candidate complexity levels, much like penalty strengths in ridge/lasso. The first component is not necessarily an unbiased or causally interpretable estimate of a biological process. A component is a weighted direction chosen by an algorithm. In the R lab, `selectNcomp()` recommends a component count from CV; it does not prove the suggested directions carry a specific biological mechanism.

18.3 Which method would you choose?

If the aim is interpretability of a prespecified exposure, start with a carefully justified regression specification rather than blindly favoring predictive dimension reduction. If the aim is risk prediction with correlated high-dimensional predictors, compare ridge, elastic net, PCR and PLS using the same outer validation design. If the sample is small and candidate dimension high, simpler models or stronger prespecification may beat flexibility. It is fully scientifically legitimate to report that the simplest model performs similarly to more complicated competitors.

19 Guided case study VI: scientific rigor in a real analysis pipeline

19.1 The difference between training, validation and deployment

Training fits the model coefficients. **Tuning** selects decisions such as the lasso penalty or number of components, ideally through inner cross-validation. **Evaluation** estimates future performance using data not used to fit or tune the candidate. **Deployment** introduces the model into a real decision process, which requires attention to calibration, transport, operational thresholds and monitoring. Results from training should never be described as external validation.

19.2 What to do with missing values during validation

Suppose one predictor has 15% missingness. If we impute it using *all* available records and the observed outcomes before forming validation folds, we pass information from the held-out sample into the training pipeline. For predictive evaluation, fit missing-data preprocessing on the training fold and apply an appropriate training-derived approach to the held-out fold without using its outcome. This often requires

more elaborate code than a stand-alone imputation function, and if that pipeline cannot be validated reliably, do not claim a leakage-free performance estimate.

19.3 A plausible reporting paragraph versus an unjustified one

Defensible: *We prespecified age, maternal weight, smoking and prior obstetric history for an exploratory continuous birth-weight regression. We assessed functional form, estimated adjusted mean differences and inspected residual diagnostics. Internal bootstrap validation quantified optimism. Predictive shrinkage methods were compared using fivefold outer cross-validation with inner penalty tuning. Results are limited by a small historical single-center sample.*

Not defensible: *Our model proved that smoking caused lower birth weight because the regression coefficient was significant; variables selected by lasso are the true biological drivers; the best training R^2 establishes clinical accuracy.*

19.4 Final integration: a question to ask before any new model

What information may legitimately be available at the time the prediction or scientific comparison is made? This single question catches much target leakage, many confounding misinterpretations and some timing errors. Follow it with: *What assumptions make this coefficient or error metric meaningful? What could make it fail in a new population?* When these questions are answered first, implementation with `rms`, `Hmisc`, and penalized methods becomes a controlled scientific workflow rather than a collection of plausible-looking functions.

PART II
R Laboratory: Fitting, Diagnosing and Validating Real Models

20 Lab 0: scientific design before code

Companion to ISLR 3.6 and ESL 3.2–3.6; supplementary implementation details.

This laboratory uses the real, publicly distributed `MASS::birthwt` teaching dataset, recorded at one medical center in 1986. The response is birth weight in *grams*, a continuous outcome appropriate for demonstrating regression. The covariates include maternal age, baseline weight, documented smoking, race classification and obstetric history. **The goal is statistical education, not clinical inference or treatment guidance.** The historical dataset is small (189 observations) and not an adequate basis for deploying a contemporary birth-weight prediction model. The lab intentionally makes the analytic steps explicit so that you can transfer them to your own appropriately governed dataset.

Two distinct target questions: (A) *Association:* how are recorded characteristics associated with the fitted conditional mean, with uncertainty? (B) *Prediction:* can a model predict previously unseen birth weights with smaller error than a simple benchmark? We will not call any association a causal effect. Our core adjustment set is prespecified as maternal age, recorded maternal weight, documented race category, smoking and prior preterm-labor history. This is a teaching specification, not a validated causal adjustment set.

Scientific caution

The data also contain `low`, an indicator defined from the outcome `bwt`. Including `low` as a predictor for `bwt` would provide information derived from the answer: classic *target leakage*. We remove it before any regression. Recode categories as factors, check impossible values, and explicitly document what is and is not measured. Recorded race labels are social/administrative classifications, not a genetic or biological mechanism.

21 Lab 1: reproducible setup, import and data dictionary

21.1 Install and load only what we need

Run the install command once. In subsequent sessions load the libraries; the remaining labs are designed to run in order in one R session. Package versions evolve, so keep a record of your actual session when reporting results.

```
# Run once; install from a trustworthy CRAN mirror.
install.packages(c("Hmisc", "rms", "MASS", "glmnet",
                   "glmnetUtils", "pls", "leaps"))

library(Hmisc)
library(rms)
```

```r
library(MASS)
library(glmnet)
library(glmnetUtils)
library(pls)

set.seed(2026)
d <- MASS::birthwt
stopifnot(nrow(d) == 189, "bwt" %in% names(d))
str(d)
```

Why these tools? `Hmisc` documents data and supports scientifically motivated imputation; `rms` packages an explicit regression modeling strategy; `glmnet` supplies the actual lasso/elastic-net solver while `glmnetUtils` exposes its formula interface; `pls` implements PCR/PLS; `leaps` supports subset selection. We will *not* compute OLS coefficients from manually constructed matrices.

21.2 Specify factor coding and remove leakage

The original dataset uses integer codes for categories. Treating `race=1,2,3` as continuous would impose a meaningless linear ordering. Likewise, smoking, hypertension and uterine irritability are indicators rather than numerical doses. Recoding helps the model estimate appropriately labeled comparisons.

```r
# The binary 'low' variable is defined from birth weight.
d$low <- NULL  # Exclude outcome-derived predictor (leakage).

d$race <- factor(d$race, levels=c(1,2,3),
                 labels=c("White","Black","Other"))
d$smoke <- factor(d$smoke, levels=c(0,1),
                  labels=c("No","Yes"))
d$ht <- factor(d$ht, levels=c(0,1),
               labels=c("No","Yes"))
d$ui <- factor(d$ui, levels=c(0,1),
               labels=c("No","Yes"))
d$ptl_any <- factor(d$ptl > 0,
                    levels=c(FALSE,TRUE), labels=c("No","Yes"))
d$ftv_cat <- cut(d$ftv, breaks=c(-Inf,0,1,Inf),
                 labels=c("0","1","2+"))
d$ptl <- NULL; d$ftv <- NULL

stopifnot(all(d$bwt > 0), all(d$age > 0),
          all(d$lwt > 0), !anyNA(d))
str(d)
```

Validate and interpret

Check: 189 rows; the outcome is numeric and still measured in grams; factor variables have meaningful labels; `low` does not appear in `names(d)`. These checks are more important than immediately fitting a regression model. **Interpretation:** recoding changes how coefficients are parameterized but not the underlying recorded observations.

22 Lab 2: descriptive statistics and exploratory analysis with Hmisc

22.1 Start with sample composition and missingness

Before modeling, examine distributions, category counts and the number of observed values for every variable. In real EHR analyses, add code-set definitions, observation windows, plausibility checks, repeated-person structure, and site-specific missingness. Here the dataset has a single row per record and initially no missing values.

```
Hmisc::describe(d)
summary(d)
table(d$race, useNA="ifany")
table(d$smoke, useNA="ifany")
sapply(d, function(z) sum(is.na(z)))

# Only numeric variables should enter this simple correlation.
Hmisc::rcorr(as.matrix(d[c("bwt","age","lwt")]),
             type="spearman")
```

Validate and interpret

Check: look for small factor categories, unexpected ranges and missingness. Spearman correlation examines a monotonic association without assuming it is linear. It is unadjusted and cannot replace the multivariable regression coefficient. A low pairwise correlation does not prove that a predictor is irrelevant after accounting for other predictors.

22.2 Plots first; model second

A scatterplot of birth weight against age may be noisy, and a boxplot by smoking category may show substantial overlap. Such plots are not tests of causation and are not sufficient for choosing the form of an adjusted model. Their job is to detect obvious data problems and help us anticipate nonlinearities or sparse groups.

```
par(mfrow=c(1,2))
plot(bwt ~ age, data=d, pch=16, cex=.65,
     xlab="Maternal age (years)", ylab="Birth weight (g)")
boxplot(bwt ~ smoke, data=d, xlab="Recorded smoking",
        ylab="Birth weight (g)")
par(mfrow=c(1,1))
```

23 Lab 3: fit, interpret and report a prespecified OLS model

ISLR 3.1–3.2; ESL 3.2.

23.1 Declare the adjustment set and fit with rms::ols

`rms::datadist()` stores typical values and observed ranges needed for standardized prediction and effect plots. The model formula states which variables are allowed to explain birth weight. `x=TRUE,y=TRUE` retains what `rms` needs for later bootstrap validation; it is not asking you to manually create a matrix.

```
dd <- rms::datadist(d)
options(datadist="dd")

formula_core <- bwt ~ age + lwt + race + smoke + ptl_any
fit_ols <- rms::ols(formula_core, data=d, x=TRUE, y=TRUE)
print(fit_ols)
coef(fit_ols)
anova(fit_ols)
summary(fit_ols)
```

Plain-language interpretation

How to read the table: each estimate is an adjusted difference in mean birth weight *in grams*, for a one-unit increase in a continuous predictor or relative to the reference category of a factor. The `anova()` table tests predictor contributions as terms; a factor with several levels uses a multi-degree-of-freedom test. Do not call a coefficient a causal effect, and do not treat a large p-value as proof of no association.

23.2 Cross-check against a familiar fitting function

`rms::ols()` implements ordinary least squares; `lm()` provides a transparent computational cross-check. Compare *fitted values*, which are invariant to benign reparameterizations, rather than relying only on printed coefficient labels.

```
fit_lm <- stats::lm(formula_core, data=d)
stopifnot(isTRUE(all.equal(
  as.numeric(predict(fit_ols)),
  as.numeric(predict(fit_lm)), tolerance=1e-6)))
summary(fit_lm)
confint(fit_lm)
```

Validate and interpret

Check: the same design, cases and contrasts should lead to matching OLS predictions. Inspect each coefficient's unit and reference level, then report estimate, interval and interpretation. **Do not:** decide that one variable is causal because its coefficient remains significant after adjustment.

24 Lab 4: obtain meaningful effect plots and contrasts

24.1 Use rms::Predict to show the model's conditional mean

Plot the fitted mean relationship against one predictor while holding others at reasonable reference or typical values from `datadist`. A ribbon/interval around the fitted mean is not an individual patient

prediction interval. Explain the comparison values when reporting the graph.

```
# Prediction/effect displays: 95% intervals for fitted means.
plot(rms::Predict(fit_ols, age, conf.int=0.95))
plot(rms::Predict(fit_ols, lwt, conf.int=0.95))
plot(rms::Predict(fit_ols, smoke, conf.int=0.95))

# Explicit adjusted contrast for a categorical predictor.
rms::contrast(fit_ols,
              list(smoke="Yes"), list(smoke="No"))
```

Validate and interpret

Check: record the covariate settings used for the predictions (typically stored adjustment values). The smoke contrast is an adjusted *difference between fitted means* and must be described in grams. If the CI is broad, the data do not tightly pin down the size of the adjusted difference.

25 Lab 5: nonlinear effects with restricted cubic splines

ISLR 3.3.2; supplementary deployment strategy with rms.

25.1 Why fit a spline rather than assume one slope everywhere?

A continuous association such as maternal age may not be a straight line throughout the observed range. `rms::rcs(age,3)` fits a smooth nonlinear term with three knots and two total regression degrees of freedom for age (one linear and one nonlinear). This is a modest illustration suited to a small sample, not evidence that nonlinear maternal-age effects truly exist here.

```
fit_spline <- rms::ols(
  bwt ~ rcs(age, 3) + lwt + race + smoke + ptl_any,
  data=d, x=TRUE, y=TRUE)

anova(fit_spline)  # Locate overall age and age nonlinearity tests.
plot(rms::Predict(fit_spline, age, conf.int=.95))

# Compare fitted means at two plausible ages.
rms::contrast(fit_spline,
              list(age=30), list(age=20))
```

Validate and interpret

Check: examine the *shape* and uncertainty of the age curve, not a single basis coefficient. A global test of age checks whether age contributes at all; a nonlinear test asks whether the extra curved component improves the fit. Both tests are model-dependent. Do not choose knot count by trying many values and publishing only the smallest p-value.

25.2 Scientific tradeoff: complexity versus stability

An extra degree of freedom costs information. With only 189 records, a very wiggly age function may fit chance fluctuations. Report the model's complexity, inspect tails where there may be few observations, and validate the complete model including spline choices. For a strong prior linear relationship, a simple linear term may be more defensible even if an exploratory curve looks interesting.

26 Lab 6: interaction modeling and conditional interpretation

ISLR 3.3.2; explanatory supplementary application.

26.1 Ask a specific effect-modification question

An interaction asks whether the association between age and birth weight differs across recorded smoking categories. Because the sample is small, this model is exploratory. Keep the main effects alongside the interaction and use plotted predictions or contrasts to interpret it.

```
fit_interaction <- rms::ols(
  bwt ~ rcs(age, 3) * smoke + lwt + race + ptl_any,
  data=d, x=TRUE, y=TRUE)

anova(fit_interaction)  # Look for pooled interaction tests.
plot(rms::Predict(fit_interaction, age, smoke,
                  conf.int=.95))
```

Validate and interpret

Check: if the two fitted age curves differ, that suggests possible *model-based effect modification*, but overlapping confidence intervals between curves are not an appropriate standalone test of interaction. Read the pooled interaction test and inspect the uncertainty. Do not describe a modeled difference as biological interaction or causal effect without a relevant study design and adequate confounding assessment.

27 Lab 7: confidence intervals versus prediction intervals

ISLR 3.2.2 and 3.3, mean and individual prediction.

The same fitted model can answer two different questions: the population mean at a given covariate profile, or a future individual's birth weight. For classical unpenalized OLS, `predict.lm()` can return a confidence interval for the mean or a wider prediction interval for an individual under normal, homoskedastic errors.

```
new_patient <- data.frame(
  age=25, lwt=130,
  race=factor("White", levels=levels(d$race)),
  smoke=factor("No", levels=levels(d$smoke)),
  ptl_any=factor("No", levels=levels(d$ptl_any)))

```

```r
predict(fit_lm, newdata=new_patient,
        interval="confidence", level=.95)
predict(fit_lm, newdata=new_patient,
        interval="prediction", level=.95)
```

Validate and interpret

Check: the individual prediction interval should be wider than the confidence interval for the conditional mean. Its nominal coverage assumes the classical model assumptions and a similar data-generating setting. If the covariate profile lies outside the observed training support, interpretation becomes extrapolative and the printed interval may be misleading.

28 Lab 8: diagnose residual patterns, leverage and influence

28.1 Begin with plots, not automatic deletion rules

We use the `lm` cross-check object for its familiar base R diagnostic plots. The fitted values are the same as the corresponding `rms::ols` model. We can inspect the residual–fitted relationship, normal Q–Q plot, Cook's distance and residual–leverage relationship.

```r
par(mfrow=c(2,2))
plot(fit_lm, which=c(1,2,4,5))
par(mfrow=c(1,1))

# Descriptive diagnostics; thresholds are flags, not deletion rules.
summary(stats::rstandard(fit_lm))
summary(stats::hatvalues(fit_lm))
summary(stats::cooks.distance(fit_lm))
flag <- which(stats::cooks.distance(fit_lm) > 4/nrow(d))
flag
if (length(flag) > 0) d[flag, c("bwt","age","lwt","smoke")]
```

Validate and interpret

Check: label what is visually happening: curvature, nonconstant spread, unusually large residuals, or influential records. `4/n` is a rough screening heuristic, not evidence of error and not a license to delete patients. Investigate data provenance and compare a predeclared sensitivity analysis only if scientifically justified.

28.2 When standard errors are not trustworthy

For iid homoskedastic errors, conventional OLS SEs are appropriate under model conditions. Heteroskedasticity can invalidate them. If this were an EHR dataset with repeated measures or site clustering, our `ols` teaching model would *not* handle that dependence automatically. Select clustered, longitudinal or mixed-effect methods as warranted; do not simply treat all repeated encounters as independent subjects.

29 Lab 9: bootstrap validation with rms

ESL 2.9 and 3.6; rms validation implementation.

29.1 Why internal validation matters in a small study

A fitted regression is evaluated on the same observations that shaped its coefficients; performance is optimistic. For a modest fixed predictor specification, bootstrap validation samples records with replacement, refits the model, and measures optimism. This is especially useful with limited sample size, where a single 80/20 train/test split leaves a very small test sample. We use `B=200` for a teaching demonstration; increase resamples when feasible and assess Monte Carlo stability.

```
set.seed(2026)
val_ols <- rms::validate(fit_ols, method="boot", B=200)
print(val_ols)

set.seed(2026)
cal_ols <- rms::calibrate(fit_ols, method="boot", B=200)
print(cal_ols)
plot(cal_ols)
```

Validate and interpret

Check: distinguish the apparent performance from optimism-corrected performance. The bootstrap estimate is *internal validation*, not validation in another hospital or era. A calibration plot for a continuous outcome examines agreement of observed and predicted means over the prediction range. Because the dataset is small, curves will be noisy.

29.2 What bootstrap validation does not repair

Bootstrap resampling cannot repair outcome leakage, an incoherent outcome definition, omitted confounding in a causal interpretation, site shifts or information that was used to choose a model before validation. If you try many alternative formulations and select one, the entire search must be included in each resample to evaluate the selected *strategy*. These `validate()` results apply to the already prespecified core model, not to every modeling choice made throughout the handbook.

30 Lab 10: missing data and multiple imputation using Hmisc

Supplementary scientific regression practice; not a claim that the original dataset contains missing data.

30.1 Create an explicitly artificial missing-data exercise

The supplied `birthwt` data are complete. To learn the tools, we deliberately obscure a small proportion of `lwt` values in a *copy*. This is a simulation of missing data, not a genuine missingness process. Do not publish the resulting coefficients as a finding or use this exercise to conclude that multiple imputation solves every missing-data problem.

```r
set.seed(2026)
d_miss <- d
missing_rows <- sample(seq_len(nrow(d_miss)), size=28)
d_miss$lwt[missing_rows] <- NA_real_
Hmisc::describe(d_miss)

# Impute missing lwt conditional on outcome and other covariates.
# nk=0 requests linear imputation transformations in this demo.
set.seed(2026)
imp <- Hmisc::aregImpute(
  ~ bwt + age + lwt + race + smoke + ptl_any,
  data=d_miss, n.impute=20, nk=0)
print(imp)
plot(imp)
```

Including the outcome `bwt` in the *imputation model* for inferential analysis is generally appropriate because it helps preserve relationships involving the outcome. This is *not* the same as including outcome-derived information as an input to a future prediction tool. For prediction validation, the distinction matters: the validation outcome must never be used to impute its own predictors. We will not combine this illustrative imputation with the cross-validation lab below because that would require imputation refitted correctly inside each training fold.

30.2 Fit the prespecified model across imputations

```r
mi_fit <- Hmisc::fit.mult.impute(
  formula_core, fitter=rms::ols,
  xtrans=imp, data=d_miss)
print(mi_fit)
summary(mi_fit)
```

Validate and interpret

Check: the pooled analysis should reflect within- and between-imputation uncertainty rather than pretending each inserted value is known. Document the number of imputations, variables in the imputation model, evidence for the missing-at-random assumption, and sensitivity to missing-not-at-random mechanisms where relevant. For formal inferential tests after MI, read the current `rms::processMI` guidance rather than using naive single-completed-data tests.

31 Lab 11: ESL 3.3 subset selection as a deliberately cautious baseline

31.1 Illustrate forward/backward selection but do not overclaim

Subset selection is part of the ESL chapter, so we demonstrate it. These are **exploratory software deployments**, not a recommended way to choose confounding covariates. Their selected p-values and confidence intervals, if treated as prespecified, are generally misleading. For a scientifically prespecified association model, keep theory-driven variables rather than letting the stepwise procedure decide them.

```r
formula_full <- bwt ~ age + lwt + race + smoke +
```

```
  ptl_any + ht + ui + ftv_cat

# Stepwise selection using AIC; exploratory only.
step_fit <- MASS::stepAIC(
  stats::lm(formula_full, data=d),
  direction="both", trace=FALSE)
formula(step_fit)
AIC(step_fit)
```

For a small, explicitly chosen collection of predictors, exhaustive subset selection is computationally possible. Here we use six single-degree-of-freedom candidates; the larger factor `race` is excluded from this illustration to avoid treating arbitrary dummy columns as independent scientific terms.

```
subset_fit <- leaps::regsubsets(
  bwt ~ age + lwt + smoke + ptl_any + ht + ui,
  data=d, nvmax=6, method="exhaustive")
subset_summary <- summary(subset_fit)
plot(subset_fit, scale="bic")
subset_summary$bic
```

Validate and interpret

Check: does the selected model look plausible and stable under modest data changes? This code is *not* a valid estimate of out-of-sample accuracy because selection occurred on all records. A fair performance analysis would repeat `stepAIC` or `regsubsets` within each outer training fold before predicting its held-out records. In particular, do not compare its training R^2 to an independently validated ridge model.

32 Lab 12: ridge regression using rms's penalized OLS

ESL 3.4.1; rms supports a ridge-type quadratic penalty for ols.

32.1 First demonstrate shrinkage, then distinguish tuning from inference

`rms::ols()` can fit a penalized linear model with a quadratic penalty. A penalty value such as 10 is illustrative: it is not prevalidated, and its numerical scale is specific to this fitting convention. `pentrace()` describes information criteria across candidate penalties, but choosing a penalty with an information criterion is not interchangeable with nested cross-validation for prediction.

```
ridge_rms <- rms::ols(formula_core, data=d,
                      penalty=10, x=TRUE, y=TRUE)
print(ridge_rms)
coef(ridge_rms)

# See effective complexity and criterion variation.
ridge_trace <- rms::pentrace(fit_ols,
                    penalty=c(0, 1, 5, 10, 25, 50))
print(ridge_trace)
```

Validate and interpret

Check: compare the scale and direction of penalized coefficients with the unpenalized ones and inspect whether shrinkage is appreciable. Do not interpret penalized coefficients as unbiased estimates of unpenalized population slopes or use naive `lm()` standard errors for this fit. This section shows **ridge-type** penalization in `rms`; **lasso is implemented separately in the next lab**.

33 Lab 13: deploy authentic lasso, ridge and elastic net

ESL 3.4.1–3.4.4; glmnet prediction implementation via glmnetUtils.

33.1 Use a formula instead of manually creating predictor matrices

`glmnet` fits penalized models efficiently but its base interface expects numeric predictors. `glmnetUtils` supplies a formula/data-frame interface, performing the required encoding internally. This is precisely what we want here: *deploy the regression method*, not spend an afternoon manually reproducing the coefficient algorithm. Standardization occurs internally; the formula interface handles categorical predictors.

```
# Continuous outcome (grams); predeclared predictors only.
set.seed(2026)
cv_lasso <- glmnetUtils::cv.glmnet(
  formula_full, data=d,
  family="gaussian", alpha=1, nfolds=5,
  type.measure="mse", standardize=TRUE,
  use.model.frame=TRUE)

plot(cv_lasso)
cv_lasso$lambda.min
cv_lasso$lambda.1se
coef(cv_lasso, s="lambda.1se")
```

The `lambda.min` choice minimizes average cross-validated error in the training sample. `lambda.1se` generally imposes more shrinkage for a simpler fit. Because the sample is small, different fold assignments may change the chosen penalty and selected predictors. Report this instability; do not convert a lasso-selected variable into a mechanistic claim.

33.2 Ridge and elastic net with exactly the same predictors

```
set.seed(2026)
cv_ridge <- glmnetUtils::cv.glmnet(
  formula_full, data=d, family="gaussian",
  alpha=0, nfolds=5, type.measure="mse",
  standardize=TRUE, use.model.frame=TRUE)

set.seed(2026)
cv_enet <- glmnetUtils::cv.glmnet(
  formula_full, data=d, family="gaussian",
  alpha=.5, nfolds=5, type.measure="mse",
```

```
  standardize=TRUE, use.model.frame=TRUE)

coef(cv_ridge, s="lambda.1se")
coef(cv_enet,  s="lambda.1se")
```

Validate and interpret

Check: ridge (`alpha=0`) typically keeps nonzero penalized coefficients; lasso (`alpha=1`) can set some to zero; elastic net lies between. **Do not compare lambda values directly across different alpha choices as if they were on identical penalty paths.** The cross-validation error from each fitted object has been used to tune that candidate; it is not an unbiased external test of whichever method you like best.

33.3 What can and cannot be inferred from lasso selection

A lasso coefficient estimates a penalty-dependent regression parameter that may be shrunken toward zero, not an ordinary unpenalized OLS effect. Default `cv.glmnet()` does not output classical post-selection confidence intervals or causal effects. If the goal is scientific estimation of a specific exposure, prespecify an appropriate adjustment set and use an inferential framework matched to the design. If the goal is prediction, evaluate the entire learning algorithm on unused observations as in Lab 15.

34 Lab 14: deploy PCR and partial least squares using pls

ESL 3.5.1–3.5.2.

34.1 Fit the two component-based alternatives

The `pls` package accepts a formula and creates the required directions internally. `scale=TRUE` makes the predictors' units comparable for deriving directions. `validation="CV"` selects components using held-out errors. In prediction applications, we must subsequently evaluate the *full procedure*, including component selection, in outer validation samples.

```
set.seed(2026)
pcr_fit <- pls::pcr(formula_full, data=d,
                    scale=TRUE, validation="CV",
                    segments=5)
set.seed(2026)
pls_fit <- pls::plsr(formula_full, data=d,
                     scale=TRUE, validation="CV",
                     segments=5)

summary(pcr_fit)
summary(pls_fit)
pls::validationplot(pcr_fit, val.type="RMSEP")
pls::validationplot(pls_fit, val.type="RMSEP")
```

34.2 Choose number of components by a documented rule

```
n_pcr <- pls::selectNcomp(pcr_fit, method="onesigma")
n_pls <- pls::selectNcomp(pls_fit, method="onesigma")
c(PCR=n_pcr, PLS=n_pls)
```

Validate and interpret

Check: record whether the rule chose zero, one or several components. Zero means the intercept-only prediction was preferred under the heuristic. Do not force at least one component simply because the analyst expected a complicated model. PCR and PLS directions are not causal factors; their clinical interpretation must be established separately.

35 Lab 15: nested validation of an entire predictive strategy

ESL 2.9, 3.4 and 3.6; supplementary reproducible performance evaluation.

35.1 Why the inner CV error is not a fair final score

`cv_lasso$cvm` selected a penalty from the available training data. If we pick whichever penalty gives the smallest inner CV error and report that same minimum as the model's generalization performance, we introduce selection optimism. A nested plan uses outer folds for honest held-out predictions, with a fresh inner CV tuning operation inside each outer training set. Here we compare a fixed full OLS formula with tuned lasso using *the same predictors*; both estimate future birth weight in grams.

```
# Outer CV assesses prediction; inner CV tunes lasso lambda.
rmse <- function(y, pred) sqrt(mean((y-pred)^2))
mae  <- function(y, pred) mean(abs(y-pred))

set.seed(2026)
K <- 5
outer <- sample(rep(seq_len(K), length.out=nrow(d)))
p_ols <- p_lasso <- rep(NA_real_, nrow(d))

for (k in seq_len(K)) {
  train <- d[outer != k, , drop=FALSE]
  test  <- d[outer == k, , drop=FALSE]

  # Fixed OLS model, fitted using only the outer training set.
  m <- stats::lm(formula_full, data=train)
  p_ols[outer == k] <- as.numeric(
    predict(m, newdata=test))

  # Penalty tuned ONLY within the corresponding training set.
  set.seed(4000 + k)
  tuned <- glmnetUtils::cv.glmnet(
    formula_full, data=train, family="gaussian",
    alpha=1, nfolds=5, type.measure="mse",
    standardize=TRUE, use.model.frame=TRUE)
  p_lasso[outer == k] <- as.numeric(
```

```
    predict(tuned, newdata=test, s="lambda.1se"))
}

stopifnot(!anyNA(p_ols), !anyNA(p_lasso))
score <- rbind(
  OLS   = c(RMSE=rmse(d$bwt,p_ols),
            MAE=mae(d$bwt,p_ols)),
  Lasso = c(RMSE=rmse(d$bwt,p_lasso),
            MAE=mae(d$bwt,p_lasso)))
print(round(score, 2))
```

Validate and interpret

Check: the outer test outcome is never passed to the lasso-fitting or tuning call. All five held-out folds are predicted once. Because there are only 189 records, results are noisy; re-running with several independent outer splits can provide a sensitivity assessment but should not become a search for a favorable split. If OLS and lasso have very similar errors, prefer the scientifically defensible simpler workflow rather than declaring a narrow winner.

35.2 A simple baseline and calibration sanity check

The mean-only baseline below is also evaluated fold by fold, with the training-set mean used to predict each outer test fold. This avoids comparing a complex algorithm against an artificially weak in-sample constant.

```
p_mean <- rep(NA_real_, nrow(d))
for (k in seq_len(K)) {
  p_mean[outer == k] <- mean(d$bwt[outer != k])
}
rbind(
  Baseline = c(RMSE=rmse(d$bwt,p_mean),
               MAE=mae(d$bwt,p_mean)),
  score)

plot(p_lasso, d$bwt, xlab="Outer-fold predictions (g)",
     ylab="Observed birth weight (g)", pch=16, cex=.7)
abline(0, 1, lty=2)
```

Validate and interpret

Check: consider whether improved RMSE is large enough to matter for the intended scientific or operational decision. A scatterplot against the identity line is a diagnostic, not external validation. A few exceptional predictions can dominate RMSE; include MAE and investigate performance across meaningful subgroups if sample sizes allow.

36 Lab 16: reporting, reproducibility and transition to your own analysis

36.1 Write a complete scientific methods paragraph

For an applied paper, a regression methods section should include: the eligible population and unit of analysis; outcome measurement, predictor definitions and timing; prespecified adjustment set; use and rationale for nonlinear terms and interactions; handling of missingness; evaluation of modeling assumptions; treatment of clustering; the exact tuning objective and CV design if regularized; uncertainty reporting; internal/external validation; software and package versions; and limitations of generalizability.

36.2 Save an audit trail

```r
# Capture the models and software provenance locally.
saveRDS(list(data_dictionary=names(d),
             formula_core=formula_core,
             formula_full=formula_full,
             ols=fit_ols,
             spline=fit_spline,
             validation=val_ols,
             nested_cv=score),
        file="linear_regression_teaching_results.rds")

writeLines(capture.output(sessionInfo()),
           "linear_regression_sessionInfo.txt")
```

Validate and interpret

Check: can someone else identify the outcome, analytic sample, coding decisions, data leakage exclusions, target model, error metric, tuning plan and software versions without asking you? Can the script be rerun from the beginning in a fresh R session? If not, the analysis is not yet reproducible.

36.3 Short decision table: which tool for which question?

Question	Primary tool(s)	Correct assessment
Adjusted mean association	`rms::ols`, `anova`, `summary`	Effect sizes, uncertainty, assumptions and design
Curved association	`rms::rcs`, `Predict`	Whole-term nonlinear test, shape, validation
Overfitting in fixed model	`rms::validate`, `calibrate`	Bootstrap optimism; not external proof
Missing predictor data	`Hmisc::aregImpute`, `fit.mult.impute`	Missingness assumptions and pooled uncertainty
Sparse prediction	`glmnetUtils::cv.glmnet` (lasso)	Nested tuning and outer error
Diffuse correlated signal	Ridge / elastic net	Same outer validation as competitors
Latent predictor directions	`pls::pcr`, `pls::plsr`	Cross-validated component count

Appendix A: six complete worked interpretation exercises

Exercise A1. A model coefficient with units

Suppose $\widehat{Y} = 2500 + 8X - 170Z$, where X is maternal age in years and $Z = 1$ indicates recorded smoking. Predict the difference in mean outcome comparing age 30 to age 20 at the same value of Z. **Solution:** the difference is $8(30 - 20) = 80$ grams. It does not depend on Z because no interaction is included. This is an adjusted association under the model, not necessarily a causal effect.

Exercise A2. Why a statistically convincing slope may have weak predictive value

Suppose a slope is 15 grams/year with an extremely small p-value but residual SD of 700 grams. **Solution:** a large enough sample may estimate the average slope precisely, yet prediction for individual records can remain very inaccurate because much outcome variation is unexplained. Statistical evidence for a nonzero mean association is not the same as clinical prediction accuracy.

Exercise A3. Compare confidence and prediction intervals

At x_0, suppose the predicted mean is 2900 grams with a 95% mean CI of (2800,3000), but the individual prediction interval is (1500,4300). **Solution:** the CI concerns the average outcome for records with the specified covariates. The prediction interval concerns one new individual and includes individual variability; it is wider. Neither interval is automatically a valid clinical thresholding tool.

Exercise A4. Interpret an interaction correctly

Suppose $\hat{Y} = 2000 + 30X - 100Z - 10XZ$. What is the estimated slope of X at $Z = 0$ and $Z = 1$? **Solution:** 30 when $Z = 0$; $30 - 10 = 20$ when $Z = 1$. The interaction coefficient -10 is the estimated difference between those two slopes. Do not infer effect modification causally without an appropriate design.

Exercise A5. Ridge or lasso for correlated predictors?

Suppose 12 biomarkers all reflect a shared biological process and most have small predictive signals. **Solution:** ridge may be a sensible candidate because it distributes shrinkage across correlated predictors; elastic net is another candidate if some sparsity is desired. Lasso may select unstable representatives of correlated groups. None can be declared best without honest validation on the target population.

Exercise A6. Why leakage invalidates a perfect model

Suppose the target is birth weight and a predictor is $\texttt{low} = \mathbf{1}(\text{birthweight} < 2500)$. A model including `low` predicts extremely well. **Solution:** the predictor is mathematically derived from the outcome, so the model is cheating by using information unavailable before knowing the outcome. Remove it, redefine predictor timing, and rerun the full training and validation workflow. More complicated algorithms will not repair leakage.

Appendix B: a practical manuscript-ready analysis checklist

1. **Research target:** Write the question as association, prediction, or causal effect. State the unit of analysis and timing of predictors relative to the outcome.
2. **Data source:** Give date ranges, eligible population, codes/measurements, inclusion/exclusion criteria and site differences. Record whether repeated records belong to the same person.
3. **Sample audit:** Describe missing data, units, impossible values, transformations, categorical coding, overlap and sampling representation.
4. **Model specification:** Separate fixed scientific predictors from exploratory options. If nonlinearities or interactions are tested, record degrees of freedom and rationale.
5. **Diagnostics:** Examine residual patterns, heteroskedasticity, dependence, leverage, influential records and extrapolation.
6. **Inference:** Report adjusted estimates with units, intervals and assumptions; avoid interpreting a selected algorithmic coefficient as causal.
7. **Prediction:** State loss function and baseline, resampling folds and grouping, penalty/component tuning, optimism correction, and external testing if available.
8. **Missingness:** State assumptions, imputation variables, number of imputations, compatibility with validation, and sensitivity analyses.
9. **Reproducibility:** Provide code, package/session versions, seed, model and tuning formulas, and enough information to rerun without discretionary post hoc choices.
10. **Limitations:** Discuss small sample size, measurement quality, shifts across institutions or time, missing causal information, and clinical implementation constraints.

Appendix C: verified chapter and software reference map

Primary uploaded textbooks

An Introduction to Statistical Learning, uploaded Chapter 3 *Linear Regression*, especially Sections 3.1–3.4 and the R lab.

The Elements of Statistical Learning, uploaded Chapter 2 *Overview of Supervised Learning* and Chapter 3 *Linear Methods for Regression*, particularly Chapter 3 Sections 3.1–3.9.

Formatting inspiration

The uploaded 48-page *Introduction to Statistical Learning for Public Health Using K-Nearest Neighbors* handout: an integrated narrative from first principles through applications, coding and methodological cautions. The present text is a new original teaching synthesis; it does not reproduce textbook passages or their copyrighted figures.

R documentation checked for the applied interfaces

- `rms::ols`: ordinary and penalized least squares; `x`, `y`, `penalty`.
- `rms` overview: `datadist`, `rcs`, `Predict`, `anova`, `contrast`.

- `rms::validate.ols` and `rms::calibrate`: internal validation and calibration.
- `Hmisc::aregImpute` and `Hmisc::fit.mult.impute`: imputation and pooled model fitting.
- `glmnetUtils::cv.glmnet`: formula-based CV for lasso and elastic net without manual predictor-matrix construction.
- `glmnet`: penalty conventions and Gaussian objective.
- `pls::selectNcomp`: component-count selection by the one-sigma rule.
- `MASS::birthwt`: data provenance, units, and source definitions.

Scientific caution

Verification status. The mathematical examples have been checked independently, and the R function calls and roles were checked against published package documentation. An R runtime was not installed in the document-generation environment, so this edition has **not** been executed line-by-line in a live R session. Before using any implementation for research, run the labs in order, inspect the actual output, record package versions, and test the complete pipeline on your properly governed data. No computed model results or performance values have been fabricated.